

Effect of consecutive repeated sprint and resistance exercise bouts on acute adaptive responses in human skeletal muscle

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Coffey VG, Jemiolo B, Edge J, Garnham AP, Trappe SW, Hawley JA. Effect of consecutive repeated sprint and resistance exercise bouts on acute adaptive responses in human skeletal muscle. *Am J Physiol Regul Integr Comp Physiol* 297: R1441–R1451, 2009. First published August 19, 2009; doi:10.1152/ajpregu.00351.2009.—We examined acute molecular responses in skeletal muscle to repeated sprint and resistance exercise bouts. Six men [age, 24.7 ± 6.3 yr; body mass, 81.6 ± 7.3 kg; peak oxygen uptake, 47 ± 9.9 ml·kg⁻¹·min⁻¹; one repetition maximum (1-RM) leg extension 92.2 ± 12.5 kg; means $\pm$ SD] were randomly assigned to trials consisting of either resistance exercise (8×5 leg extension, 80% 1-RM) followed by repeated sprints (10×6 s, 0.75 N·m torque·kg⁻¹) or vice-versa. Muscle biopsies from vastus lateralis were obtained at rest, 15 min after each exercise bout, and following 3-h recovery to determine early signaling and mRNA responses. There was divergent exercise order-dependent phosphorylation of p70 S6K (S6K). Specifically, initial resistance exercise increased S6K phosphorylation ($\sim 75\%$ $P < 0.05$), but there was no effect when resistance exercise was undertaken after sprints. Exercise decreased IGF-I mRNA following 3-h recovery ($\sim 50\%$, $P = 0.06$) independent of order, while muscle RING finger mRNA was elevated with a moderate exercise order effect ($P < 0.01$). When resistance exercise was followed by repeated sprints PGC-1 α mRNA was increased (REX1-SPR2; $P = 0.02$) with a modest distinction between exercise orders. Repeated sprints may promote acute interference on resistance exercise responses by attenuating translation initiation signaling and exacerbating ubiquitin ligase expression. Indeed, repeated sprints appear to generate the overriding acute exercise-induced response when undertaking concurrent repeated sprint and resistance exercise. Accordingly, we suggest that sprint-activities are isolated from resistance training and that adequate recovery time is considered within periodized training plans that incorporate these divergent exercise modes.

repeated sprint ability; resistance training; adaptation

CONCURRENT TRAINING CAN BE defined as the incorporation of two or more divergent exercise modes in a periodized training program. Since the classic work of Hickson (30), studies of concurrent training have attempted to determine the interaction of endurance and resistance exercise on subsequent adaptation in skeletal muscle. Given the distinct phenotypes generated in response to either chronic endurance or resistance training and the apparent incompatibility of the exercise-induced adaptation (fatigue resistance vs. high-force production), previous work in this area has endeavored to elucidate the potential interference

effect of concurrent training. In this regard, results of research investigating adaptation and performance changes, when concurrent training is undertaken, has been equivocal (39, 40). While the specific physiological effects of concurrent training in skeletal muscle are far from established, work from our laboratory has revealed divergent responses for several molecular pathways when athletes undertake acute concurrent training (15). Indeed, we have shown that the proximity of endurance and resistance exercise reduced the extent of the desired molecular response and did not promote optimal activation of pathways to promote training specificity for each exercise mode.

An integral part of training programs for individuals participating in many sporting activities is repeated sprint ability. However, we currently lack information on the molecular training responses in skeletal muscle that occur in response to this type of activity. Sprint training promotes alterations in ion regulation and buffering capacity with little or no increase in muscle size (4, 21, 49). Moreover, high-intensity sprint exercise results in a large ionic disturbance, including the accumulation of muscle lactate and hydrogen ions (H⁺). Of note is that sprint exercise has recently been shown to induce a molecular adaptive profile associated with mitochondrial biogenesis and increased capacity for glucose and fatty acid oxidation (23). To the best of our knowledge, the molecular interactions of concurrent resistance and repeated sprint training have not been investigated. Indeed, such an approach represents a novel expansion of the conventional concurrent training paradigm. Consequently, we currently have no understanding of the potential mechanisms that might underlie exercise-induced adaptation and enhanced/impaired muscle function with concurrent resistance and repeated sprint training. Despite the potential similarity in the work-to-recovery ratio between sprint and resistance exercise, the greater rate of force production and inadequate recovery during repeated sprints generates large disturbance to ion homeostasis, the accumulation of lactate, and altered substrate metabolism during ensuing repetitions that characterize a distinct exercise overload. (7). Thus, in contrast to the resistance exercise-induced increases in translation initiation and protein synthesis (65), repeated sprints might be expected to inhibit anabolic processes. Accordingly, the aim of the current study was to quantify the acute cellular responses in skeletal muscle when resistance and sprint training sessions were performed successively and to examine the effect of alternate exercise order (i.e., an initial bout of heavy resistance exercise, then repeated sprint exercise or vice versa). Given the paucity of data regarding the molec-

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ular response following repeated sprints in skeletal muscle, we examined signaling associated with the translational machinery (Akt-mTOR-p70S6K) and changes in mRNA abundance related to anabolic growth [insulin-like growth factor (IGF)], proteasome-dependent protein breakdown [muscle RING finger (MuRF), atrogin], and mitochondrial biogenesis/carbohydrate metabolism [peroxisome proliferator-activated receptor- γ coactivator-1 α (PGC-1 α), hexokinase] (14). We also specifically chose this suite of molecular markers to compare the responses to repeated sprints with our previous work showing concurrent resistance and endurance exercise-induced changes during the acute postexercise recovery period (15). We hypothesized that the different exercise modes undertaken in close proximity would amplify the potential incompatibility of the acute molecular profile for each form of contractile activity.

METHODS

Subjects

Six male subjects [age, 24.7 ± 6.3 yr; body mass, 81.6 ± 7.3 kg; peak oxygen uptake ($\dot{V}O_{2\text{ peak}}$), 47 ± 9.9 ml $\cdot$ kg $^{-1}\cdot$ min $^{-1}$; leg extension one repetition maximum (1-RM), 92.2 ± 12.5 kg (means $\pm$ SD)] who had been participating in regular concurrent resistance and endurance training ($>3 \times$ per wk) volunteered for this study. The experimental procedures and possible risks associated with the study were explained to each subject who gave written informed consent prior to participation. This study was approved by the Human Research Ethics Committee of the Royal Melbourne Institute of Technology, Melbourne, Australia.

Study Design

The study employed a randomized cross-over design where each subject completed two acute concurrent repeated sprint and resistance exercise sessions separated by a 2-wk recovery period, during which time subjects were instructed to continue their habitual physical activity. One experimental trial consisted of consecutive resistance and then repeated sprint exercise bouts, while the other trial was performed using the reverse exercise order, i.e., repeated sprint then resistance exercise.

Preliminary Testing

$\dot{V}O_{2\text{ peak}}$, $\dot{V}O_{2\text{ peak}}$ was determined during an incremental test to exhaustion on a Lode cycle ergometer (Groningen, The Netherlands). The protocol has been described in detail previously (29). In brief, subjects commenced cycling at a workload equivalent to 2 W/kg for 150 s. Thereafter, the workload was increased by 25 W every 150 s until volitional fatigue, defined as the inability to maintain a cadence > 70 rev/min. Throughout the test, subjects breathed through a mouthpiece attached to a metabolic cart (Parvomedics, Sandy, Utah) that was calibrated before testing using a 3-liter Hans-Rudolph syringe and gases of known concentration (4% CO $_2$ and 16% O $_2$).

Maximal strength. The strength of quadriceps was determined during a series of single repetitions on a plate-loaded leg extension machine until the maximum load lifted was established (1-RM). Repetitions were separated by 3-min recovery and were used to establish the maximum load/weight that could be moved through the full range of motion once, but not a second time. Exercise range of motion was 85 degrees, with leg extension endpoint set at -5 degrees from full extension.

30-s Wingate. Wingate testing was undertaken on a Lode Excalibur Sport cycle ergometer (Groningen, The Netherlands). Subjects remained seated throughout the protocol that commenced with 60-s cycling at 75 W immediately followed by 30-s maximal sprint cycling against a resistance equal to 0.7 N $\cdot$ m torque $^{-1}\cdot$ kg body mass $^{-1}$.

Diet/Exercise Control

Before an experimental trial (described below), subjects refrained from training and other vigorous physical activity for a minimum of 48 h. Subjects were provided with standardized prepacked meals that consisted of 3 g CHO/kg body mass, 0.5 g protein/kg body mass, and 0.3 g fat/kg body mass consumed as the final caloric intake the evening prior to reporting for an exercise testing session.

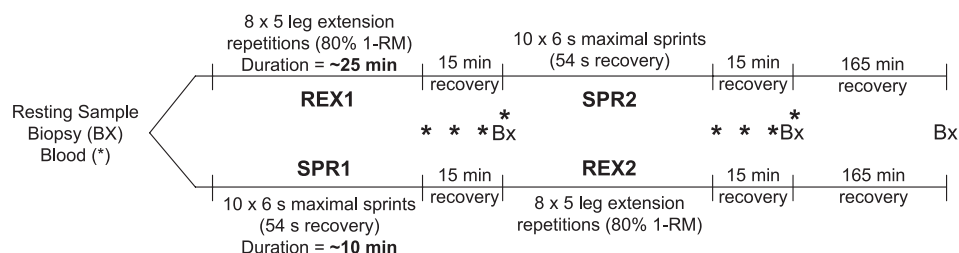
Experimental Trials

An overview of the study testing protocol is shown in Fig. 1. On the morning of an experimental trial, subjects reported to the laboratory after an ~ 10 -h overnight fast. After resting in the supine position for ~ 15 min, local anesthesia (2–3 ml of 1% Xylocaine) was administered to the skin, subcutaneous tissue, and fascia of the vastus lateralis of the subject's leg in preparation for the series of muscle biopsies. A resting (basal) biopsy was taken using a 5-mm Bergstrom needle modified with suction. Approximately 150 mg of muscle was removed and blotted to remove excess blood; ~ 15 –20 mg was then cut and placed in RNALater (Ambion, Austin, TX), and the remainder was immediately frozen in liquid nitrogen. Subjects then completed the two exercise sessions (described in detail below). Fifteen minutes after completion of the first exercise bout, a second biopsy was taken. Subjects then performed the second exercise bout. After 15 min of recovery from the second exercise session, a third muscle biopsy was taken, and subjects rested in the supine position until a fourth muscle biopsy was taken following the 3-h recovery period. Each muscle biopsy was taken from a separate site, and all samples were stored at -80°C until subsequent analysis. In addition, blood samples (~ 2 ml) were taken at rest preexercise, 0, 5, 10, and 15 min after the first exercise bout and 0, 5, 10, 15, and 30 min after the second exercise bout.

Resistance Exercise

Following a standardized warm-up (2 $\times$ 5 repetitions at 50% and 60% 1-RM, respectively), subjects performed eight sets of five repetitions at 80% 1-RM. Each set was separated by a 3-min recovery period during which the subject remained seated on the leg extension machine. Contractions were performed at a set metronome cadence approximately equal to 30 degrees/s and strong verbal encouragement was provided during each set.

Fig. 1. Schematic of the experimental trials incorporating consecutive bouts of diverse exercise. This study utilized a randomized cross-over design where subjects completed both exercise orders [1 repetition maximum (1-RM)]. The duration of each exercise bout was ~ 25 min vs. ~ 10 min for resistance (REX) and repeated sprint (SPR) exercise, respectively.



Repeated Sprint Exercise

Subjects undertook 10 × 6-s maximal sprint efforts at a constant load. Subjects performed 10 maximal effort sprint cycling repetitions of 6-s duration against 0.75 N·m torque⁻¹·kg⁻¹ body mass on a Lode Excalibur Sport cycle ergometer. The initial sprint repetition was preceded by 60 s cycling at 75 W, and subjects were instructed to initiate each sprint at 80 rpm and remain seated throughout the sprint exercise. Each sprint repetition was separated by 49-s passive rest followed by a 5-s rolling start for each subsequent sprint (i.e., each sprint started at 1-min intervals).

Analytical Procedures

Blood/muscle lactate and pH/H⁺. Whole blood was immediately analyzed for lactate concentration using an automated glucose/lactate analyzer (model YSI 2300; Yellow Springs, Ohio). Approximately 25 mg of frozen muscle was freeze dried and blood/connective tissue was removed from the powdered muscle tissue. For determining muscle pH, samples (~3 mg) were homogenized on ice for 2 min in a solution containing sodium fluoride (10 mM NaF) at a dilution of 30 mg dry muscle/ml of homogenizing solution. The muscle homogenate was then placed in a circulating water bath at 37°C for 5 min prior to and during the measurement of pH. The pH measurements were made with a microelectrode (model MI-415; Microelectrodes, Bedford, NH) connected to a pH meter (model SA 520; Orion Research, Cambridge, MA). For determining muscle lactate, samples (~4 mg) were enzymatically assayed by using the spectrophotometry methods of Harris et al. (28).

Western blot analysis. Muscle samples were homogenized in buffer containing 50 mM Tris·HCl, pH 7.5, 1 mM EDTA, 1 mM EGTA, 10% glycerol, 1% Triton X-100, 50 mM NaF, 5 mM Na pyrophosphate, 1 mM DTT, 10 µg/ml trypsin inhibitor, 2 µg/ml aprotinin, 1 mM benzamide, and 1 mM PMSF. After determination of protein concentration (Pierce, Rockford, IL), lysate was resuspended in Laemmli sample buffer, separated by SDS-PAGE, and transferred to polyvinylidene fluoride membranes blocked with 5% nonfat milk, washed with TBST (10 mM Tris·HCl, 100 mM NaCl, 0.02% Tween 20) and incubated with primary antibody (1:1,000) overnight at 4°C. Membranes were incubated with secondary antibody (1:2,000), and proteins were detected via chemiluminescence (Amersham Biosciences, Buckinghamshire, UK; Pierce Biotechnology, Rockford, IL) and quantified by densitometry. Sample (50 µg) time points for each subject were run on the same gel and quantified relative to α-tubulin protein abundance. Polyclonal antiphospho-mammalian target of rapamycin (mTOR)^{ser2448} (cat. no. 2971); and monoclonal antiphospho-Akt^{ser473} (cat. no. 9271), -tuberin (TSC2)^{thr1462} (cat. no. 3617), -S6 ribosomal protein^{ser235/6} (cat. no. 4856) were from Cell

Signaling Technology (Danvers, MA). Antiphospho-AMPK^{thr172} was raised against AMPK-α peptide (KDGEFLRpTSCGAPNY) as described previously (13) and anti-phospho-p70 S6K^{thr389} was from Millipore (cat. no. 04-392; Temecula, CA). Monoclonal anti-α-tubulin control protein antibody (cat. no. T6074) was from Sigma-Aldrich (St Louis, MO).

Total RNA extraction and RNA quality check. Total RNA (23.1 ± 1.28 mg of tissue) was extracted in TRI reagent (Molecular Research Center, Cincinnati, OH). The quality and integrity (RIN of 8.56 ± 0.18 SE) of extracted RNA (0.28 µg/ mg of tissue) was evaluated using an RNA 6000 Nano LabChip kit on an Agilent 2100 Bioanalyzer.

RT and real-time PCR. Oligo(dT) primed first-strand cDNA was synthesized (150 ng total RNA) using SuperScript II RT (Invitrogen, Carlsbad, CA). Quantification of mRNA (in duplicate) was performed in a 72-well Rotor-Gene 3000 Centrifugal Real-Time Cycler (Corbett Research, Mortlake, NSW, Australia). The current study was limited in that, to the best of our knowledge, no previous work has determined the suitability or otherwise of selected housekeeping genes following repeated sprint exercise. We selected GAPDH based on a validation study from our laboratory that has shown GAPDH to be most stable in all human skeletal muscle fiber types following acute exercise (36, 66). All primers used in this study were mRNA specific (on different exons and over an intron) and were designed for SYBR Green chemistry using Vector NTI Advance 9 software (Invitrogen) [Table 1; sequences published elsewhere: PGC-1α (27), antrogin-1, muscle RING finger, forkhead box O 3a (Foxo3a) (43), and myogenic differentiation factor (MyoD) (56)]. A melting curve analysis was generated to validate that only one product was present. The details regarding RT and PCR reaction parameters have been reported previously (43, 56).

Relative qPCR data analysis. Relative gene expression analysis comparing expression of a gene-of-interest in relation to a reference-gene, based on the distinct cycle (Ct) differences, the ΔCt model (42), was used in this study as described in detail previously (56, 57, 68).

Statistical Analysis

We calculated sample size using the methods of Hopkins (32), and we have previously reported statistical significance for related physiological variables in cross-over studies employing six to eight subjects (15–17, 52). All data were analyzed by two-way repeated-measures ANOVA (2-factor: time × exercise order) with Student-Newman-Kuels post hoc analysis. Statistical significance was established when *P* < 0.05 (SigmaStat for windows Version 3.11). Lacking any information on the smallest substantial changes in acute responses to divergent exercise, we performed additional post hoc analysis in which each individual's data were back transformed

Table 1. Summary of mRNA primers for genes of interest used in PCR reactions

Name	Sequence	GeneBank No.	Amplicon Size, bp	Annealing Temperature, °C	mRNA Region, bp
IGF-1					
Sense	5'-CAGGGGCTTTTATTTCAACAAGCCCA-3'	NM_000618	127	59	419–545
Antisense	5'-TGCAGCAATACATCTCCAGCCTCCTTA-3'				
MGF (IGF-IEc)					
Sense	5'-CAGAAGTATCAGCCCCCATCTACCAACAA-3'	NM_001111283	91	59	616–706
Antisense	5'-TGCACTCCCTCTACTTGCGTTCTTCAA-3'				
TFAM					
Sense	5'-GGAAACCAAAAAGACCTCGTTCAGCTT-3'	NM_003201	86	59	589–674
Antisense	5'-TTTTCTGCGGTGAATCACCGTT-3'				
HK II					
Sense	5'-CAGCAGAACAGCCTGGACGAGAGCAT-3'	MN_000189	126	60	3745–3870
Antisense	5'-GTCAAACCTCTCTCGCCGGTGGAT-3'				
CS					
Sense	5'-CATGGACTGGCAAATCAGGAAGTGCTT-3'	MN_004077	144	58	1091–1220
Antisense	5'-CTGGAACAACCGCTCTGAGTTGAG-3'				

Sense, forward primer; anti-sense, reverse primer; IGF-I, insulin-like growth factor 1; MGF, mechano-growth factor; TFAM, mitochondrial transcription factor A; HK II, hexokinase II; CS, citrate synthase.

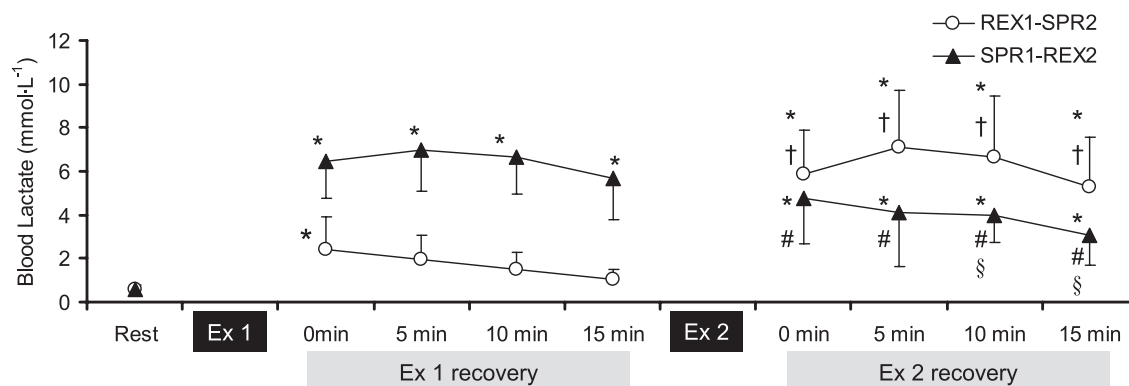


Fig. 2. Blood lactate at rest preexercise, immediately and at 5-min intervals during 15-min postexercise recovery following each individual exercise bout (Ex 1 and 2) incorporating REX (8×5 leg extensions at 80% 1-RM) and SPR (10×6 s maximal effort) in alternate exercise orders. Results are group means $\pm$ SD. Significant difference ($P < 0.05$) vs. *rest, †SPR2 vs. REX1, #REX2 vs. REX1, and §REX2 vs. SPR1.

against the mean of their corresponding resting preexercise values. For further details of the statistical methods employed for magnitude-based inferences the reader is referred to Hopkins et al. (33). Briefly, log-transformed delta values between data time points were directly compared and converted to Cohen effect sizes (ES) to determine the magnitude of change for each exercise bout and alternate exercise order (31). We chose a default confidence interval of 90% to calculate ES via a spreadsheet, making the same assumptions about sampling distributions that statistical packages use to derive P values (33). We interpreted the magnitude of the ES by using conventional threshold values of 0.2 as the smallest ES, 0.5 as a moderate ES, and 0.8 as a large ES (18, 33). Data are expressed as arbitrary units $\pm$ SD.

RESULTS

Repeated Sprint Performance

There was a significant main effect for mean power output within the different sprint repetition bouts ($P < 0.01$). Moreover, mean power declined $\sim 11\%$ (sprint 1–10) when repeated sprints were undertaken after resistance exercise (REX1-SPR2, 965 ± 140 W vs. 860 ± 106 W) and $\sim 19\%$ when sprints preceded resistance exercise (SPR1-REX2, 990 ± 194 W vs. 804 ± 76 W). However, mean power generated during the entire set of 10×6 s sprints was similar regardless of exercise order, with a significant difference between exercise orders only within SPR4 ($P < 0.05$).

Blood/Muscle Lactate and pH/ H^+

Changes in postexercise blood lactate concentration resulted in differences for time and exercise order, and significant time $\times$ exercise order interactions ($P < 0.01$; Fig. 2). Specifically, when resistance exercise preceded repeated sprints, blood lactate levels were only elevated above rest immediately after REX1 (0 min, $\sim 2.4 \pm 1.5$ mmol/l, $P < 0.05$; Fig. 2). In

contrast, blood lactate concentration was significantly elevated above rest at every time point throughout the 15-min recovery when resistance exercise was undertaken after repeated sprints (REX2 ~ 3 – 4.7 mmol/l, $P < 0.05$; Fig. 2). In addition, postexercise blood lactate concentration was consistently increased above rest following repeated sprints regardless of exercise order ($P < 0.05$; Fig. 2).

There were significant main effects among the levels of time for muscle lactate concentration (Table 2). Repeated sprint exercise significantly increased muscle lactate above resting levels regardless of exercise order ($P < 0.05$), and lactate concentration was higher following SPR1 compared with REX2 ($P < 0.05$). Moreover, muscle lactate was only significantly elevated above rest following resistance exercise when preceded by repeated sprints (rest vs. REX2, $P < 0.05$; Table 2). There were also significant main effects among the levels of time for both pH (rest vs. REX, rest vs. SPR, $P < 0.05$) and H^+ (rest vs. SPR, $P < 0.05$; Table 2), but there were no significant interactions. The decrease in pH and increase in H^+ was greatest when sprints were undertaken prior to resistance exercise compared with sprints performed subsequent to resistance exercise (Table 2).

Signaling Responses

Main effects for Akt^{ser473} phosphorylation verged on significance for the different levels of time ($P = 0.055$). The initial bout of exercise had little effect on phosphorylation of Akt regardless of contraction mode (Fig. 3A). However, increased Akt phosphorylation 15 min after resistance exercise was undertaken, subsequent to repeated sprints (SPR1-REX2), resulted in a large disparity compared with initial resistance exercise (REX2 vs. REX1, $\sim 180\%$; ES > 1.0 ; Fig. 3A).

Table 2. Muscle lactate (La) and H^+ content, and pH measured at rest and 15 min postexercise following each exercise bout in the alternate exercise orders

	Rest	REX1	SPR2	Rest	SPR1	REX2
La, mmol/kg dry wt	9.2 ± 4	15.8 ± 4	$28.2 \pm 16^*$	8.3 ± 4.6	$42.5 \pm 19^{*\dagger}$	$25.5 \pm 21^*$
pH	7.06 ± 0.09	6.98 ± 0.13	6.95 ± 0.20	7.06 ± 0.09	$6.92 \pm 0.15^*$	6.99 ± 0.12
H^+ , mmol/kg dry wt	89 ± 18	109 ± 36	123 ± 63	88 ± 16	125 ± 40	105 ± 27

Values are means $\pm$ SD. REX, resistance exercise; SPR, repeated sprints. Significantly different ($P < 0.05$) vs. *rest, vs. †REX2.

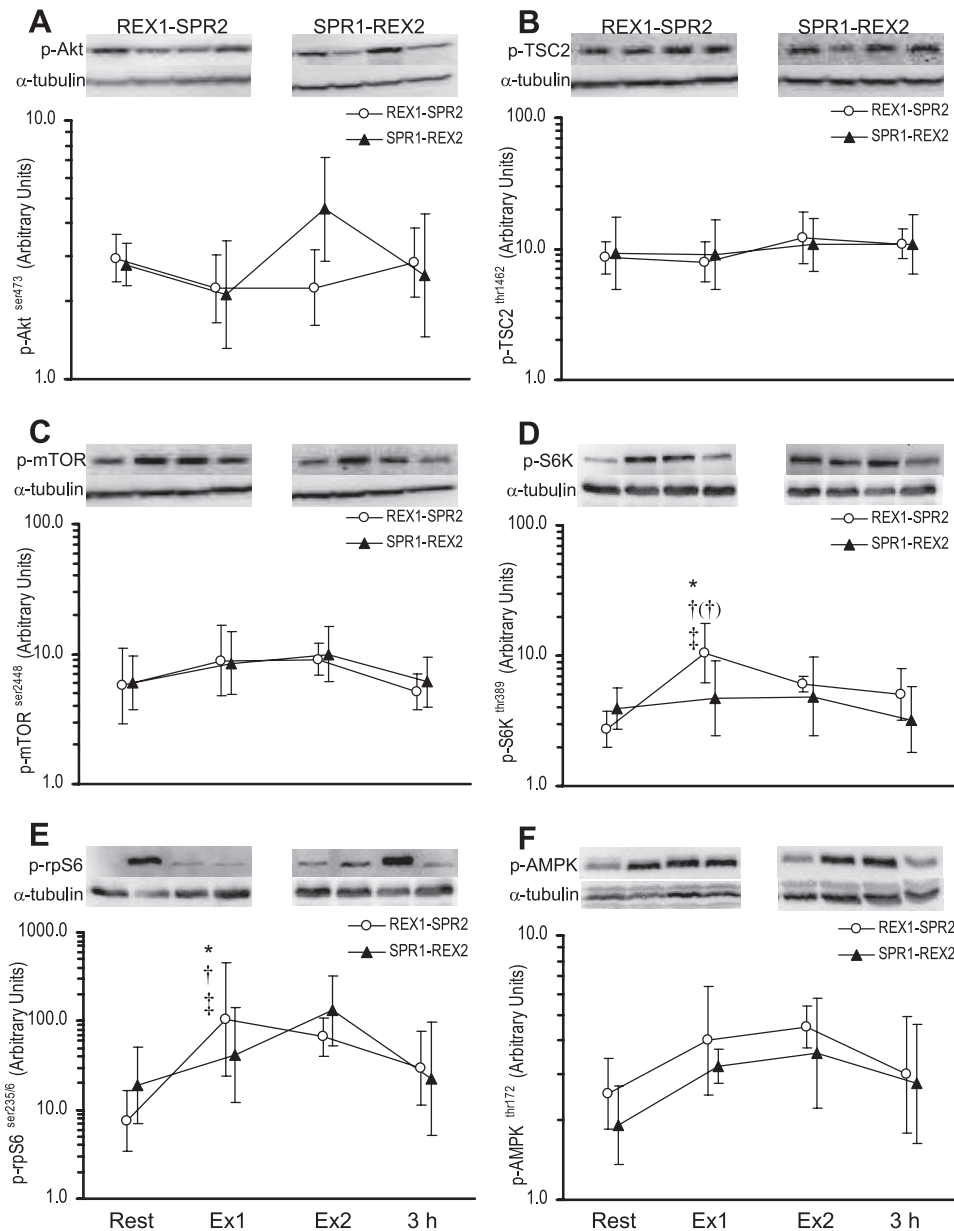


Fig. 3. Phosphorylated Akt^{ser473} (A), tuberin^{thr1462} (TSC2; B), mammalian target of rapamycin^{ser2448} (mTOR; C), p70 S6K^{thr389} (D), S6 ribosomal protein^{ser235/6} (p-rpS6; E), and adenosine monophosphate-activated kinase α ^{thr172} (AMPK; F) relative to α -tubulin at rest preexercise, 15 min after each individual exercise bout (Ex 1 and 2), and 3 h after cessation of training. Subjects completed 2 experimental trials incorporating consecutive REX (8 $\times$ 5 leg extensions at 80% 1-RM) and SPR (10 $\times$ 6 s maximal effort) bouts performed in alternate order. Results are group means $\pm$ SD, and data are log-transformed values as arbitrary units. Significant difference ($P < 0.05$) vs. *rest, †SPR2, (‡)REX2, and ‡3 h.

Phosphorylation of Akt was largely unaffected 15 min after each bout of repeated sprints, and phosphorylation state returned to resting levels 3 h following cessation of contractile activity, regardless of exercise order.

Changes in TSC2^{thr1462} phosphorylation in response to resistance and repeated sprint exercise were modest (Fig. 3B). There were minor changes to TSC2 phosphorylation following an initial bout of contractile activity regardless of mode, while a moderate increase was observed with the successive exercise bout (REX2 vs. REX1, $\sim 28\%$; ES, 0.5; SPR2 vs. SPR1, $\sim 56\%$; ES, 0.8; Fig. 3B). There was a significant main effect for mTOR^{ser2448} phosphorylation 15 min following resistance exercise ($P < 0.05$), while elevated phosphorylation above rest after repeated sprints approached significance ($P = 0.056$; Fig. 3C), but there were no significant interactions. Of note, there were corresponding mTOR phosphorylation responses to contractile activity regardless of exercise mode (Fig. 3C). Elevated

TSC2 and mTOR phosphorylation abated and returned to resting levels following 3-h recovery from the consecutive exercise bouts.

Diverse contractile activity and alternating exercise order induced divergent p70 S6K^{thr389} phosphorylation responses ($P < 0.05$; Fig. 3D). Specifically, an initial bout of resistance exercise generated an increase in S6K phosphorylation that was different from all other time points within the REX1-SPR2 exercise order ($P < 0.05$; Fig. 3D). Moreover, repeated sprints did not promote phosphorylation of S6K, and there was no effect when resistance exercise was undertaken after sprints, resulting in a significant order effect within resistance exercise bouts (REX1 vs. REX2, $\sim 74\%$; ES > 1.0 , $P < 0.05$). There was also disparity in Δ S6K phosphorylation from rest following the 3-h recovery period, with elevated phosphorylation observed when resistance exercise preceded repeated sprints (REX1-SPR2, $\sim 56\%$; ES > 1.0 ; Fig. 3D). Changes in ribo-

somal protein (rp) S6^{ser235/6} phosphorylation resulted in a significant main effect within the different levels of time ($P < 0.05$; Fig. 3E). The enhanced phosphorylation of rpS6 15 min after initial resistance exercise was significantly different from rest and other postexercise time points (REX1-SPR2, $P < 0.05$; Fig. 3E). Of note, prior repeated sprints did not prevent increased rpS6 phosphorylation with subsequent resistance exercise, but the magnitude of elevation from rest was moderately lower (REX1 vs. REX2, $\sim 50\%$; ES, 0.66). After 3-h recovery, rpS6 phosphorylation abated but remained higher above rest following the resistance exercise-repeated sprints exercise order (REX1-SPR2, $\sim 69\%$; ES > 1.0 ; Fig. 3E).

Changes in AMPK^{thr172} phosphorylation resulted in a significant main effect within the different levels of time ($P < 0.05$; Fig. 3F). There were similar increases in AMPK phosphorylation 15 min after an initial bout of contractile activity regardless of exercise mode. Moreover, the increase in phosphorylation of AMPK was sustained following the subsequent exercise bouts with no disparity in the magnitude of change from rest (Fig. 3F). Following 3-h recovery, AMPK phosphorylation was moderately higher above rest when repeated sprints preceded resistance exercise (SPR1-REX2, $\sim 22\%$; ES, 0.53; Fig. 3F).

mRNA Responses

IGF-I-mechano-growth factor-MyoD. Due to limitations in tissue sample size, the mRNA analysis was restricted to comparisons at rest and 3 h after cessation of contractile activity. Comparable decreases in IGF-I mRNA from rest following 3-h recovery from exercise generated a large magnitude of effect (ES ~ 1.0) that verged on significance ($\sim 50\%$, $P = 0.06$; Fig. 4A). However, there was little difference in the magnitude of effect between the alternate exercise orders ($\sim 16\%$; ES, 0.2). Similarly, there was a moderate decrease in mechano-growth factor (MGF) mRNA abundance 3 h postexercise that was not significantly different from rest ($\sim 50\%$; ES, ~ 0.7 , $P = 0.08$), with negligible disparity in the exercise order effect ($\sim 28\%$; ES, 0.26; Fig. 4B). The elevation in MyoD mRNA 3 h after the cessation of contractile activity was large (ES > 1.0) but failed to reach significance and was similar regardless of exercise order ($\sim 300\%$, $P = 0.06$; Fig. 4C).

Atrogin-MuRF-Foxo3a. Variation in individual responses resulted in little change in atrogin and Foxo3a (Foxo3a not shown) mRNA abundance following the consecutive exercise bouts, with little discrepancy in the effect of the different exercise orders (ES < 0.2 ; Fig. 5A). In contrast, the mRNA abundance of MuRF was significantly elevated above rest following 3-h recovery from resistance exercise undertaken prior to repeated sprints, and also after sprints were performed before resistance exercise (ES > 1.0 , $P < 0.01$; Fig. 5B). The magnitude of increase in MuRF mRNA abundance was moderately exacerbated when repeated sprints were undertaken subsequent to resistance exercise (REX1-SPR2 vs. SPR1-REX2, $\sim 25\%$; ES, 0.75; Fig. 5B).

PGC-1 α -hexokinase II-mitochondrial transcription factor A-citrate synthase. There was a significant main effect for the levels of time for PGC-1 α mRNA in response to the combined exercise bouts ($P = 0.04$; Fig. 6A). The increased abundance of PGC-1 α mRNA, when repeated sprints preceded resistance exercise, failed to reach significance despite a large

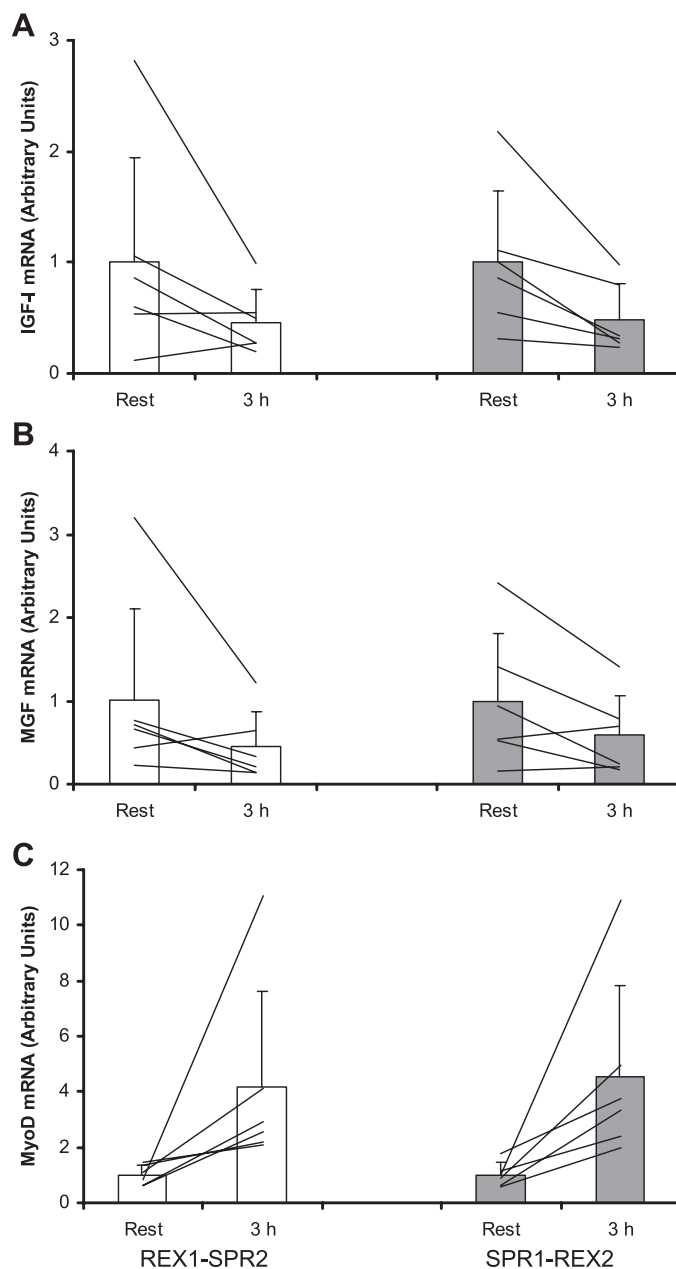


Fig. 4. Insulin-like growth factor 1 (IGF-I; A), mechano-growth factor (MGF; B), and myogenic differentiation factor (MyoD; C) mRNA abundance at rest preexercise and 3 h after cessation of training. Subjects completed 2 experimental trials incorporating consecutive REX (8×5 leg extensions at 80% 1-RM) and SPR (10×6 s maximal effort) bouts performed in alternate order. Results are individual responses and group means in arbitrary units.

effect (SPR1-REX2, ES > 1.0 , $P = 0.09$), while the resistance exercise followed by repeated sprints exercise order was significantly different from rest (REX1-SPR2, ES > 1.0 , $P = 0.02$). There was a moderate distinction in the relative increase above resting levels with augmented PGC-1 α mRNA when repeated sprints followed resistance exercise (REX1-SPR2, $\sim 35\%$; ES, 0.46; Fig. 6A). Hexokinase II mRNA abundance was enhanced in response to contractile activity and was significantly elevated above rest after both exercise orders (ES > 1.0 , $P < 0.05$; Fig. 6B). The magnitude of increase in hexokinase II mRNA 3 h postexercise was moderately higher

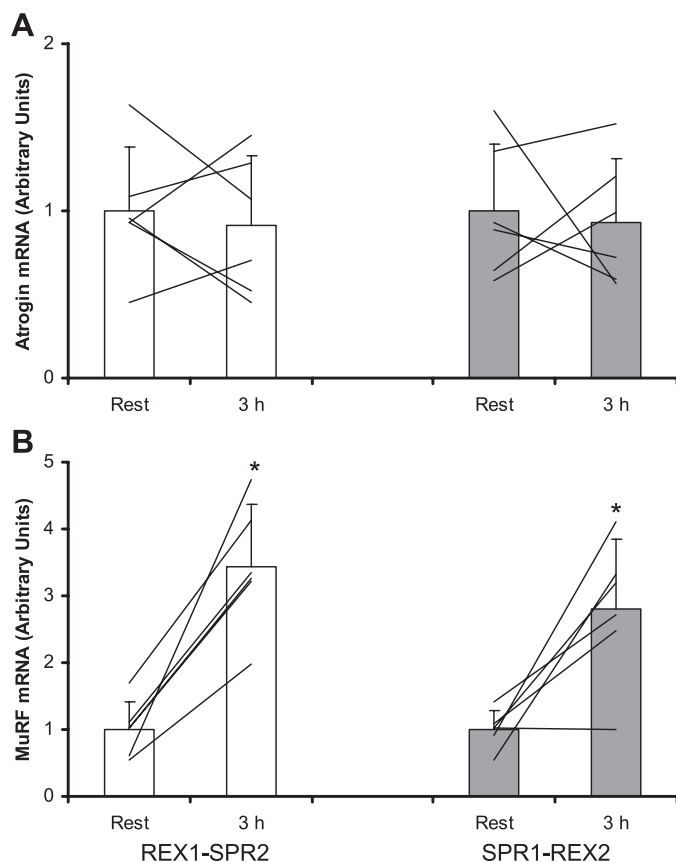


Fig. 5. Atrogin (A) and muscle-ring finger (MuRF; B) mRNA abundance at rest preexercise and 3 h after cessation of training. Subjects completed 2 experimental trials incorporating consecutive REX (8×5 leg extensions at 80% 1-RM) and SPR (10×6 s maximal effort) bouts performed in alternate order. Results are individual responses and group means in arbitrary units. *Significant difference ($P < 0.05$) vs. rest.

when repeated sprints were undertaken before resistance exercise (SPR1-REX2, ~46%; ES, 0.4; Fig. 6B). There was disparity in individual mitochondrial transcription factor A and citrate synthase mRNA responses (data not shown) culminating in minor changes in mean mRNA abundance from rest following the 3-h recovery period.

DISCUSSION

Individual training sessions provide the overload stimulus that generates specific adaptive events that ultimately alter cellular characteristics for enhanced performance capacity in skeletal muscle. We have recently established adaptive responses for several cellular and molecular pathways when athletes undertake consecutive acute resistance and endurance (concurrent) training (15). In the present study, we have taken a novel approach to the concurrent training paradigm by combining resistance exercise with repeated sprint performance to examine the acute specificity of training adaptation and exercise-order effects in skeletal muscle. Specifically, we show attenuated S6K and rpS6 phosphorylation following repeated sprints in an exercise order-dependent manner. Moreover, repeated sprints also appeared to diminish the anabolic response to resistance exercise highlighted by a decrease in IGF-I mRNA and concomitant increase in MuRF mRNA abundance following 3-h recovery. In addition, we observed

robust increases in the mRNA abundance of PGC-1 α and hexokinase with only a modest exercise order distinction, highlighting the capacity of high-intensity sprint exercise to stimulate acute mitochondrial and metabolic adaptive responses.

There were minimal differences in the decline of mean power despite a greater decrement with initial repeated sprints, likely due to higher initial power outputs and accompanying metabolic perturbations when undertaking sprints first (i.e., without any metabolic activity or residual fatigue from prior contractile activity) generating a slightly greater fatigue index (8). Moreover, during the 15-min recovery period following each repeated sprint bout, blood lactate concentration was similar (Fig. 2). However, the effect of elevated blood lactate levels following initial repeated sprint exercise resulted in higher concentrations after subsequent REX2. This finding demonstrates that circulating blood lactate remained elevated during resistance exercise undertaken after sprints (Fig. 2). Muscle lactate was only significantly elevated above rest following resistance exercise when preceded by repeated sprints (Table 2). Differences in muscle H^+ content were less clear, but the largest changes were observed following an initial bout of repeated sprints (Table 2). It may be reasonable to conclude that resistance exercise performed after the repeated sprint

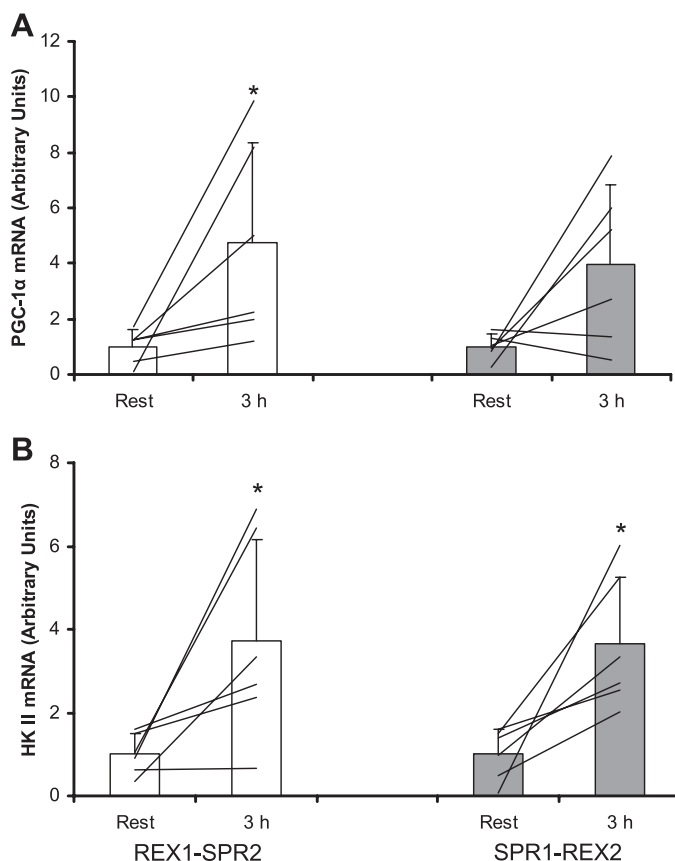


Fig. 6. Peroxisome proliferator-activated receptor- γ coactivator-1 α (PGC-1 α ; A) and hexokinase II (HKII; B) mRNA abundance at rest preexercise and 3 h after cessation of training. Subjects completed two experimental trials incorporating consecutive REX (8×5 leg extensions at 80% 1-RM) and SPR (10×6 s maximal effort) bouts performed in alternate order. Results are individual responses and group means in arbitrary units. *Significant difference ($P < 0.05$) vs. rest.

repetitions was undertaken in the presence of greater intramuscular and whole body metabolic acidosis compared with the initial resistance exercise bout. These findings provide support for the notion that the different exercise modes generate divergent stimuli that promote dissimilar signaling activity and gene expression. Thus, the close proximity to resistance exercise and metabolic changes as a consequence of the repeated sprints is significant given its potential to create an incompatible environment for resistance exercise-induced acute responses within the muscle milieu.

Akt has been characterized as a critical junction for multiple cell processes including glucose transport and hypertrophy in skeletal muscle (10, 64). In the present study, changes in Akt-TSC2-mTOR and AMPK phosphorylation and any subsequent exercise-order effect were modest. Following 15-min recovery from the initial exercise bout there was little divergence in Akt^{ser473} phosphorylation between exercise modes. In contrast, when resistance exercise was undertaken after sprints, we observed elevated phosphorylation of Akt 15 min postexercise, resulting in a disparity in the magnitude of effect from rest ($ES > 1.0$; Fig. 3A). To the best of our knowledge, this is the first study to examine the insulin/IGF signaling pathway in skeletal muscle following short, repeated sprint activity. Accordingly, we present new data indicating that repeated sprints do not appear to promote Akt-TSC2-mediated kinase activity. In support of this observation, Gibala et al. (23) have reported a decrease in Akt phosphorylation immediately (0 min) after 4×30 -s all-out cycling bouts, each separated by 4-min recovery. However, it should be noted that the acute postexercise cell-signaling time course, following both resistance and sprint exercise is currently unknown. Whether the disparity in Akt phosphorylation after a subsequent bout of resistance exercise in the present study represents the cumulative effect of the combined exercise bouts or is related to the time course of activation, remains to be established.

mTOR is a key regulatory protein for cell processes including translation and protein synthesis, and mitochondrial function (6, 19, 60). In the present study, there was a moderate increase in mTOR^{ser2448} phosphorylation from rest following the initial exercise bouts that was sustained with subsequent exercise, regardless of mode and order (Fig. 3C). The putative evidence for an Akt-independent mechanism capable of regulating mTOR phosphorylation in response to mechanical stretch (34, 35) may provide some rationale for the modest increases (main effects) in exercise-induced phosphorylation of mTOR without concomitant changes in Akt phosphorylation. We also observed similar changes in AMPK^{Thr172} phosphorylation independent of exercise order (Fig. 3F). Given the potential for high-intensity contraction during repeated sprints to induce rapid changes in energy turnover and alter the ATP-to-AMP ratio in skeletal muscle, a significant increase in AMPK activation might be expected. Indeed, Chen et al. (12) and Gibala et al. (23) have shown elevated AMPK activity immediately after 30-s sprints. The results of the present study indicate that the 15-min recovery time point may not have coincided with peak AMPK activity following the divergent contractile activity. Although few previous studies have measured both AMPK and mTOR phosphorylation/activity in vivo human skeletal muscle, Dreyer et al. (20) have previously shown a significant increase in both AMPK activity and mTOR phosphorylation following a bout of resistance training. Mas-

cher et al. (45) have also examined Akt-mTOR signaling after 60 min of cycling and observed increased mTOR phosphorylation 0 and 30 min postexercise, a time when AMPK activity is also elevated. In addition, work in a transgenic animal model by McGee et al. (47) provides support for the role of AMPK in promoting, rather than inhibiting, skeletal muscle growth independent of its role in metabolism. Clearly, much work remains to elucidate the interactions and subsequent physiological function of these signaling proteins in human skeletal muscle.

A novel finding of the present study was the divergence in S6K^{Thr389} and rpS6^{Ser235/6} phosphorylation following the diverse exercise modes. The S6K and rpS6 proteins are proposed to regulate translational efficiency of mRNAs with a 5'-terminal oligopyrimidine tract (59), although a causative relationship remains contentious (59), and increased phosphorylation of S6K and rpS6 is associated with the adaptive response to resistance exercise (17, 20, 22, 24). The mechanisms regulating these proteins are complex, and Akt and mitogen-activated protein kinase-mediated signaling pathways may converge at rpS6 (58). Nonetheless, we (15) and others (5, 44) have also shown increased S6K/rpS6 phosphorylation after a single bout of endurance exercise in human skeletal muscle. In contrast, Gibala et al. (23) reported no change in S6K phosphorylation following multiple 30 s sprints. We provide new data indicating that short, repeated sprint activity (10×6 s) does not promote translation initiation signaling in the early recovery period, suggesting that the overriding acute response necessitates enhanced ion transport and buffering capacity (21, 49). The suppression of S6K/rpS6 phosphorylation when comparing resistance exercise undertaken after repeated sprints with initial resistance exercise is intriguing. In previous work from our laboratory we failed to see a significant exercise order effect and observed increased S6K and rpS6 phosphorylation with consecutive endurance and resistance exercise bouts (15). Of note, metabolic acidosis has been shown to decrease protein synthesis in skeletal muscle of rodents and humans (11, 38). Thus, the elevated muscle lactate and low pH immediately prior to resistance exercise when undertaken after the initial repeated sprint bout implicates local acidosis as an inhibitor of resistance exercise-induced S6K and rpS6 phosphorylation (Table 2). Collectively, these results provide support for the putative capacity of an initial bout of repeated sprints to attenuate subsequent translation initiation up to ~ 3 h postexercise, due at least in part to changes in the ionic state of the muscle milieu.

We also examined mRNA responses of select genes associated with hypertrophy, atrophy, and metabolism in skeletal muscle (14) to provide an adaptive profile generated by the consecutive bouts of diverse exercise. Changes in mRNA abundance for genes of interest implicated in regulation of hypertrophy and atrophy did not reveal any obvious exercise order effect but the dual exercise bouts appeared to inhibit the IGF-I mRNA response and promote mRNA abundance of MuRF 3 h after cessation of contractile activity (Figs. 4 and 5). A potential limitation of the present study was that muscle samples for equivalent 3-h postexercise recovery time points for each individual exercise bout were not taken. Nonetheless, our findings represent an adaptation "snapshot" 3 h after the cessation of contractile activity that characterizes the cumulative effect of the combined divergent stimuli.

IGF-I is a major growth factor that induces anabolic effects in skeletal muscle (3, 51, 62) and increased IGF-I expression is closely associated with muscle hypertrophy (1, 2). While recent work by Spangenburg et al. (63) offers putative evidence for IGF-I independent muscle hypertrophy, elevated mRNA abundance of IGF-I variants (IGF-IEa and MGF) corresponds with increases in muscle cross-sectional area following chronic exercise training (53). Few acute exercise studies have examined IGF-I mRNA and a range of IGF-I responses have been observed between 2–4 h recovery following resistance exercise (26, 48, 55). We have previously shown that following concurrent endurance and resistance exercise, IGF-IEa and MGF mRNA abundance was only attenuated when cycling immediately preceded resistance exercise (15). In contrast, the results of the present study revealed a decrease in IGF-I mRNA 3 h after the consecutive repeated sprint and resistance exercise bouts that was similar regardless of exercise order (Fig. 4A and B). Thus, we suggest that repeated sprints may have a greater negative effect on growth factor mRNA abundance compared with an acute endurance exercise bout, and this effect is mediated by proximity per se rather than exercise order when combined with resistance exercise. Gene expression and resultant synthesis of new functional protein is regulated at numerous molecular “check-points”. While strong correlations between variance in mRNA and protein levels have been observed, the possibility exists that within the complexity of adaptation in skeletal muscle changes in mRNA and protein may not always equate (41). However, more work is needed to clarify the acute effect of exercise-induced contraction and concurrent training on IGF-I mRNA responses.

We also observed similar mRNA changes in the MyoD in an exercise order-independent manner (Fig. 4C). Yang et al. (67) have shown corresponding increases in skeletal muscle MyoD mRNA abundance following both endurance and resistance exercise. Likewise, previous work from our laboratory shows that the cumulative effect of endurance and resistance exercise generates similar increases in MyoD mRNA abundance (15). Therefore, enhanced MyoD mRNA expression appears to be induced by a variety of contractile activity suggesting a minimal role in the specificity of training adaptation.

There were contrasting effects of the acute repeated sprint and resistance exercise bouts on the mRNA abundance of mediators of skeletal muscle inflammation and proteolytic responses in skeletal muscle (37, 61). Specifically, the cumulative effect of exercise induced a pronounced increase in MuRF mRNA, while variation in the individual response resulted in no alteration in mean atrogen mRNA abundance (Fig. 5, A and B). There was only a moderate disparity in the increase above rest in MuRF mRNA with the alternate exercise orders (SPR1-REX2 vs. REX1-SPR2; ES, 0.75). Atrogen and MuRF1 are E3 ubiquitin ligases implicated in skeletal muscle proteolysis via covalent tagging of specific proteins with ubiquitin for their subsequent degradation by the 26S proteasome (9, 25). We have seen a similar discrepancy in an exercise order-dependent manner which showed an increased magnitude in the MuRF mRNA response when an endurance exercise bout was performed following resistance exercise (15). Work by Louis et al. (43) indicates that MuRF mRNA abundance may be elevated 1–4 h after 30 min (75% $\dot{V}O_{2\max}$) of running and three sets of 10 repetition (70% 1-RM) resistance exercise. Thus, the immediacy of repeated sprints in the present study did not attenuate the

increase in markers of exercise-induced inflammation and may actually accentuate the muscle damage response following heavy resistance exercise. However, there is a paucity of data on the effect of repeated sprints on mRNA expression and acute concurrent training interactions, and more work is needed to elucidate the effect on anabolic and catabolic adaptation responses in skeletal muscle.

PGC-1 α has been described as a master controller of mitochondrial biogenesis, and increases in PGC-1 α mRNA have been clearly evident following an acute bout of aerobic endurance exercise (16, 46, 50, 54). We have previously shown that an acute bout of concurrent endurance and resistance exercise incorporating 30 min of cycling (70% $\dot{V}O_{2\text{ peak}}$) is insufficient to induce a robust increase in PGC-1 α mRNA abundance, regardless of exercise order. In contrast, the results of the present study reveal a significant increase in PGC-1 α mRNA despite the proximity of the divergent exercise bouts and short cumulative duration of the sprint activity (60 s). Gibala et al. (23) previously showed only a small dose of intense exercise was sufficient to elevate PGC-1 α mRNA abundance. Likewise, our data highlight the capacity for repeated bouts of supra-maximal exercise to enhance acute PGC-1 α mRNA expression despite prior resistance exercise that may promote increased PGC-1 α protein content. Indeed, this unexpected revelation was achieved utilizing exercise of half the duration (60 s) to that of Gibala et al. (23). In addition, there was a modest disparity in the exercise order effect on PGC-1 α mRNA, indicating that resistance exercise undertaken after repeated sprints may induce mild attenuation of the mitochondrial adaptive response, but such an effect remains to be clearly established. Hexokinase mRNA abundance was increased 3 h after completing the alternate exercise bouts with only a modest order effect, demonstrating a comparable metabolic adaptive response of pathways regulating energy provision. Taken together, these results strongly suggest repeated sprints are capable of stimulating an initial acute response that may ultimately enhance skeletal muscle oxidative and metabolic capacities when undertaken concurrently with resistance exercise regardless of exercise order.

Perspectives and Significance

This is the first study to examine the potential interference or additive effect of acute concurrent training incorporating repeated sprint activity and resistance exercise in skeletal muscle. We chose to employ successive exercise bouts with short recovery based on anecdotal reports from coaches of elite athletes regarding selected training regimens in a number of sports and also for the potential for time-efficient adaptation with acute concurrent training sessions for health and fitness. A novel finding of the present study is that repeated sprints appear to promote greater acute interference on markers of resistance exercise adaptation than previously observed with endurance exercise (15). Specifically, we observed an attenuated phosphorylation of kinases proximal to translation initiation and growth factor mRNA abundance. These acute responses may represent a temporary suppression of molecular processes that promote net synthesis or maintenance of muscle mass. Moreover, we also show elevated mRNA for a primary marker of inflammation/proteolysis that would exacerbate the apparent negative effect on muscle mass, and these effects

were largely evident, regardless of the order in which the consecutive exercise bouts were undertaken. An additional finding of our study was that repeated sprints may provide a concentrated stimulus for mitochondrial biogenesis and highlights the potential for short, intense exercise to promote aerobic adaptation responses. Collectively, our results indicate repeated sprint activity generates the prevailing exercise-induced adaptive response and subsequently alters the acute adaptive profile in skeletal muscle when undertaking concurrent repeated sprint and resistance exercise. Accordingly, we suggest that repeated sprint activities are isolated from resistance training and that adequate recovery time is considered within periodized training plans that incorporate these divergent exercise modes.

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